
ASSIGNMENT 1

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Question: 5.1

1 ADALINE (Adaptive Linear Neuron)

1.1 Theory

1.1.1 a) ADALINE and the Widrow-Hoff Learning Rule

ADALINE (Adaptive Linear Neuron) was introduced by Bernard Widrow and Ted Hoff in 1960 as one of the earliest neural network models. Unlike the perceptron, ADALINE uses a **linear activation function** during training and optimizes the weights using the **Least Mean Squares (LMS)** algorithm.

Given an input vector

$$x = [x_1, x_2, \dots, x_d]^T$$

the ADALINE output is

$$\hat{y} = w^T x + b$$

where w is the weight vector and b is the bias.

The learning objective is to minimize the mean squared error:

$$L = \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The Widrow-Hoff learning rule updates weights using gradient descent:

$$w^{(t+1)} = w^{(t)} + \eta(y - \hat{y})x$$

$$b^{(t+1)} = b^{(t)} + \eta(y - \hat{y})$$

where η is the learning rate.

Algorithm (ADALINE Training):

1. Initialize weights w and bias b .
2. For each training sample:
 - (a) Compute output: $\hat{y} = w^T x + b$
 - (b) Compute error: $e = y - \hat{y}$
 - (c) Update weights:

$$w \leftarrow w + \eta e x$$

- (d) Update bias:

$$b \leftarrow b + \eta e$$

3. Repeat until convergence or maximum iterations reached.

1.1.2 b) Convergence to Global Least Squares Solution

The ADALINE model minimizes the convex objective function

$$J(w) = \frac{1}{2} \|y - Xw\|^2$$

where X is the data matrix.

Since this function is a quadratic convex function in w , it has a unique global minimum given by the normal equation:

$$w^* = (X^T X)^{-1} X^T y$$

Gradient descent applied to a convex quadratic function is guaranteed to converge to this global optimum (for sufficiently small learning rate).

Unlike the perceptron, ADALINE does not rely on data being linearly separable because it minimizes squared error rather than classification mistakes. Therefore the algorithm converges even when the dataset is not separable.

Thus ADALINE has a strong convergence guarantee to the global least-squares solution.